German Property Investment Evaluator

A concise report on the ingestion, evaluation, chat, and user interface workflows.

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Doc:https://docs.google.com/document/d/1yQkWEOfsf86qoGmPRHZcrhNnO1sNauJejrdB3JJdzNc/edit?usp=sharing

Demo: https://www.loom.com/share/c6504e4e578340dd9cdb69f9fbda5953

Github Link:https://github.com/anshulgoyl/week-2-german-property-evaluator

1. Project overview

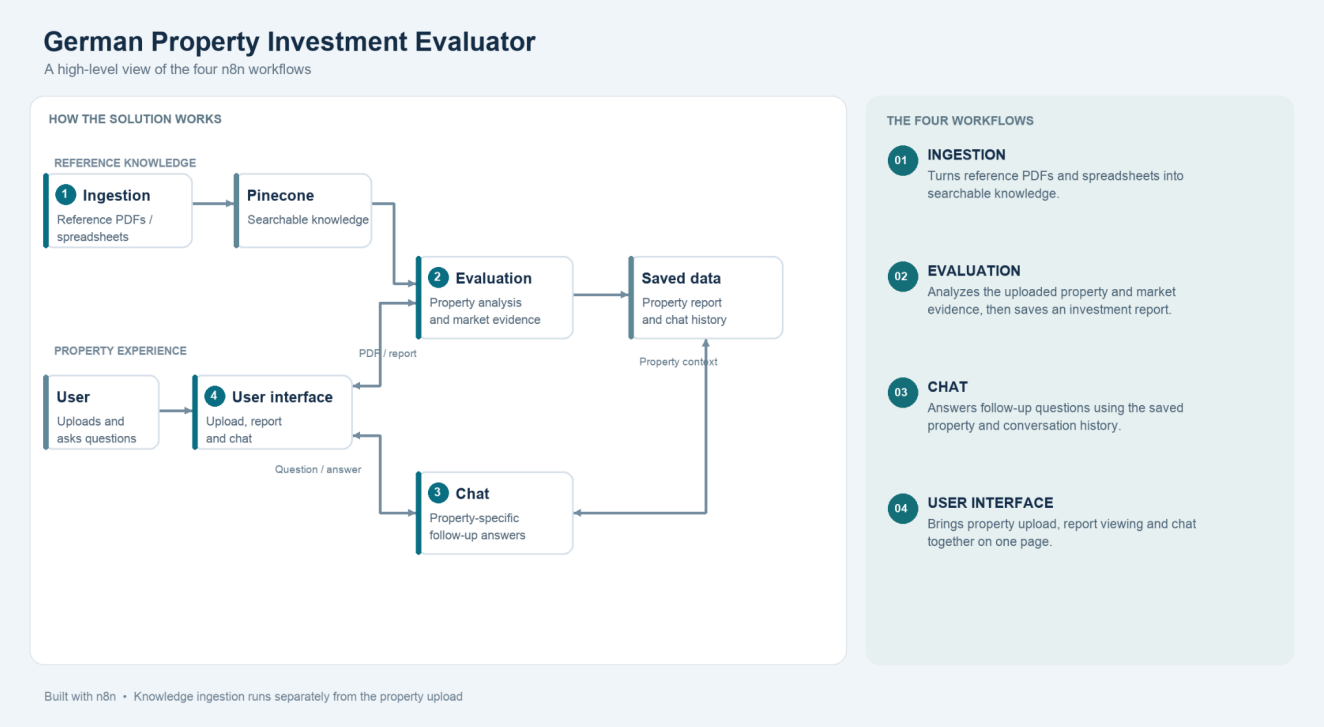
Reviewing a property for investment requires collecting listing details, assessing risks, and understanding the asking price in its market context. This project brings those steps into an initial AI-assisted screening workflow. A user uploads a property PDF, receives an investment report, and can ask follow-up questions about the same property. Reference documents are prepared separately to support the evaluation.

A. Problem Statement and framework

As I explored property investment in Germany, I found the process time-consuming and often lacking transparency. Assessing an opportunity meant bringing together property documents, market information and financial assumptions, while accounting for commissions and other purchase costs. It was also difficult to find reliable comparable properties and historical transaction prices, making it harder to judge whether an asking price was reasonable. At the same time, property investment can offer significant tax benefits, which need to be considered alongside costs and risks. I developed this solution to make my initial assessment more consistent. I can upload a property PDF, receive an investment report and ask follow-up questions about the opportunity.

Field	Description
Use case	A property investor asks, “How attractive is this German property, and what are its main risks?” The answer appears as an investment report and follow-up chat in an n8n-hosted browser interface.
Corpus	The project’s reference folder contains 13 PDFs and spreadsheets covering investment guidance, financial models and sample listings, primarily in German and English. The project owner manages the reference files; their presence in the folder does not confirm that all have been indexed.
Ingestion + cleaning	Reference files are uploaded through an n8n form and routed to PDF or Excel extraction. JavaScript prepares extracted text or spreadsheet rows for indexing; the workflow does not explicitly implement a general boilerplate-removal step.
Ingestion + freshness	Reference files are uploaded manually, and the ingestion workflow creates or updates their records in Pinecone. No automatic refresh cadence or freshness SLA is configured.
Chunking + embedding	PDF text is split into fixed 1,000-character chunks with 200-character overlap; Excel data is prepared as individual row records. Qwen3-Embedding-8B, accessed through Nebius, converts these into embeddings for semantic search.
Retrieve	The evaluator performs dense vector search in Pinecone’s `documents` namespace and requests up to 8 matches. The active chat path uses the saved property record and conversation history; its separate Pinecone retrieval nodes are disconnected.

B. The solution at a glance



Scope: The output is a starting point for research, not a certified valuation or a substitute for financial, legal, or technical due diligence.

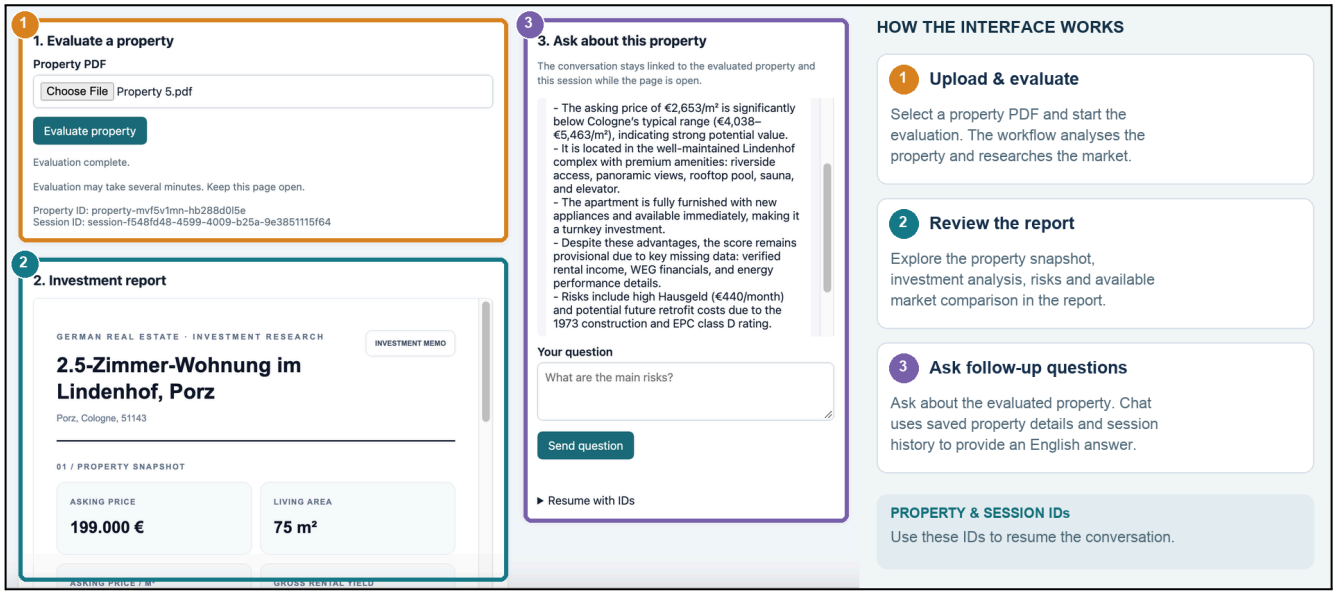
C. High Level Approach

The project applies ideas illustrated in the provided solution kits: separate ingestion and question-answering workflows, grounded document analysis, clear system diagrams, and practical testing. These examples informed the report structure; the implementation reflects the project's own workflows and constraints.

Decision	Purpose and trade-off
n8n workflows	Separate the major tasks and make the process visible; the workflows still depend on reliable handoffs and external services.
Pinecone + embeddings	Make reference material searchable by meaning; index configuration and document quality affect retrieval.
Tavily market search	Bring current search observations into the report; snippets and listing details require verification.
n8n data tables	Retain property records and conversation history; the tables require access controls and ongoing management.
Webhook-based UI	Provide a quick unified interface; sandbox, origin, and deployment settings must be configured carefully.

D. Inputs Outputs

The unified interface brings the property evaluation process into one place. Users upload a property PDF, review the generated investment report and ask follow-up questions about the property. The property and session IDs keep the conversation linked to the evaluation and allow it to be resumed later.



E. Testing and observed performance

The live flow was tested with a property PDF: the evaluation returned a property ID and report, and a follow-up question returned an answer linked to that property and session. A German-language chat question was also tested after adding an English-response instruction. These examples demonstrate the flow, but they are not a comprehensive quality evaluation. More properties and questions are needed to test extraction accuracy, source grounding, and isolation between properties. Four complete evaluations before optimization took approximately 31, 39, 43, and 65 seconds. Two later tests took 20 and 103 seconds. The slower test spent most of its time in external model calls. These small samples show substantial variability and do not establish a guaranteed improvement.

2. Data Used

The solution draws on property investment guidance, financial models and sample listings to support its analysis. For each evaluation, it also uses the uploaded property PDF and market information found through web searches.

Source	Description
Apartment investment checklist	Guidance on selecting and assessing an investment apartment
Property due-diligence checklist	A comprehensive checklist for reviewing a residential investment
Investment playbook	A Düsseldorf buy-to-let strategy and decision process

Cash-flow worksheet	A template for recording property income and expenses
Property viewing checklist	Points to inspect during a property visit
Purchase due-diligence checklist	Legal, financial and documentation checks for a purchase
Location analysis template	Factors for assessing a property's neighbourhood and wider area
Comparable valuation checklist	Guidance for comparing a property with similar properties
Property returns calculator	A spreadsheet for returns, financing and tax calculations
Investment calculation model	Property assumptions and a ten-year cash-flow projection
Investment economics calculator	Cash flow, assets and financial indicators for a rental property
Sample property listings	Listing details such as location, price and property characteristics
Scored property listings	Listings assessed against price, condition and city benchmark assumptions

3. Tools Used

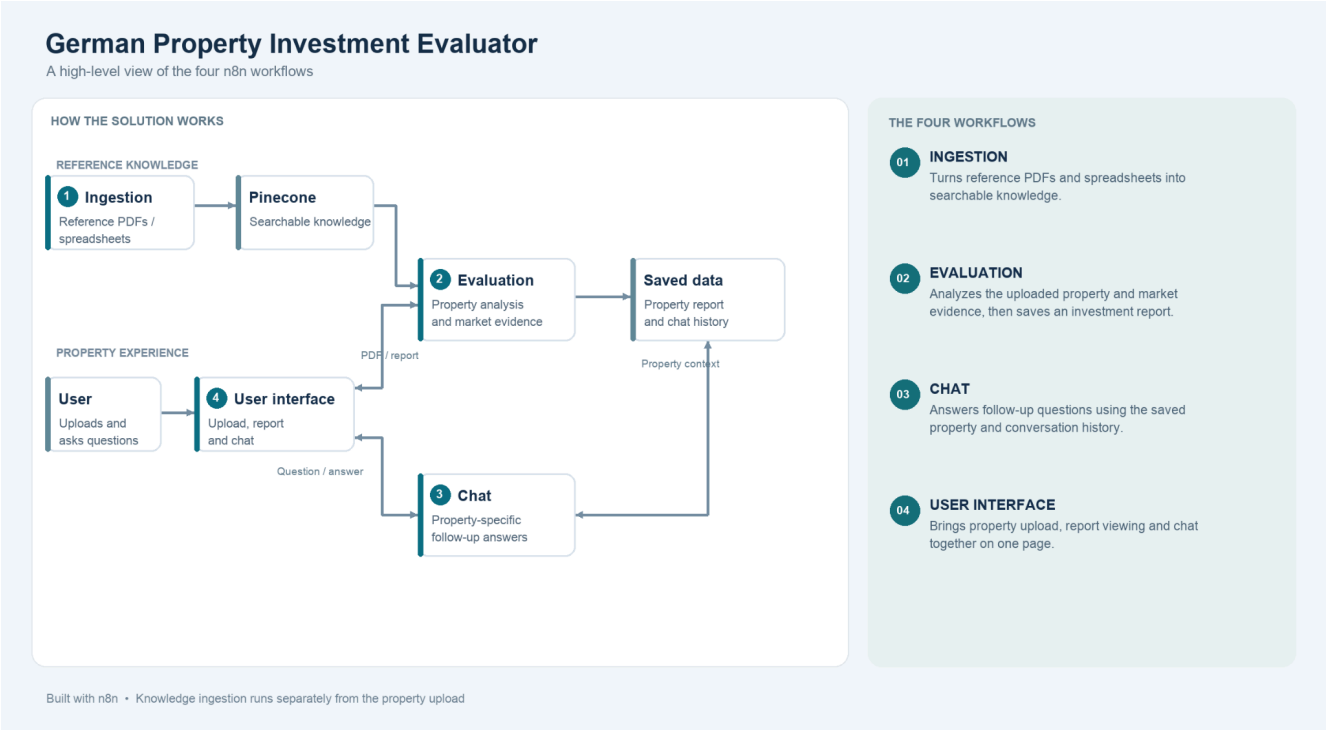
The solution combines workflow automation, AI models, a searchable knowledge base and web research to evaluate properties and answer follow-up questions.

Tool or model	**Description**
n8n	Runs the four workflows and coordinates the services. Its built-in nodes extract content from PDFs and spreadsheets.
Nebius AI	Provides API access to the AI models used in the workflows.
Qwen3-Embedding-8B	Converts reference documents and property information into numerical representations for searching the knowledge base.
Qwen3-235B-A22B-Instruct-2507	Generates investment analysis, extracts market comparables and answers follow-up questions.
Pinecone	Stores searchable reference knowledge and retrieves relevant context for property evaluation.
Tavily	Searches the web for market information and comparable property listings.

n8n data tables	Store property reports and conversation history for follow-up questions.
HTML and JavaScript	Provide the browser interface and handle data preparation and report formatting.
Webhooks and HTTP requests	Connect the interface, workflows and external services.

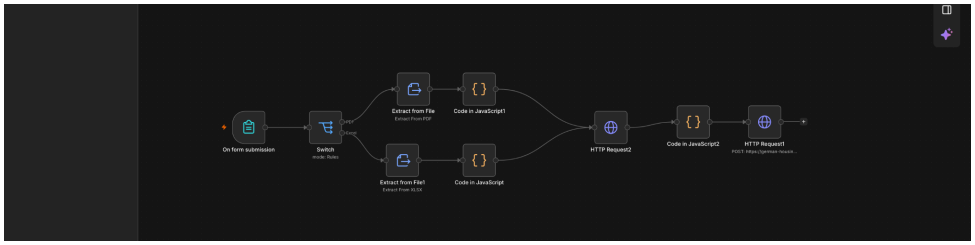
4. Workflows and deep-dives

Each workflow has a distinct role. Together they move information from reference documents and a property listing to a report and a property-specific conversation. Below is a clear description of 4 key workflows in the application.



1 Knowledge ingestion

Input: reference PDFs or spreadsheets. The workflow extracts content, divides it into chunks, creates embeddings with Nebius, and stores vectors and metadata in Pinecone. This prepares background knowledge for retrieval during evaluation. Ingestion is separate from each property upload.

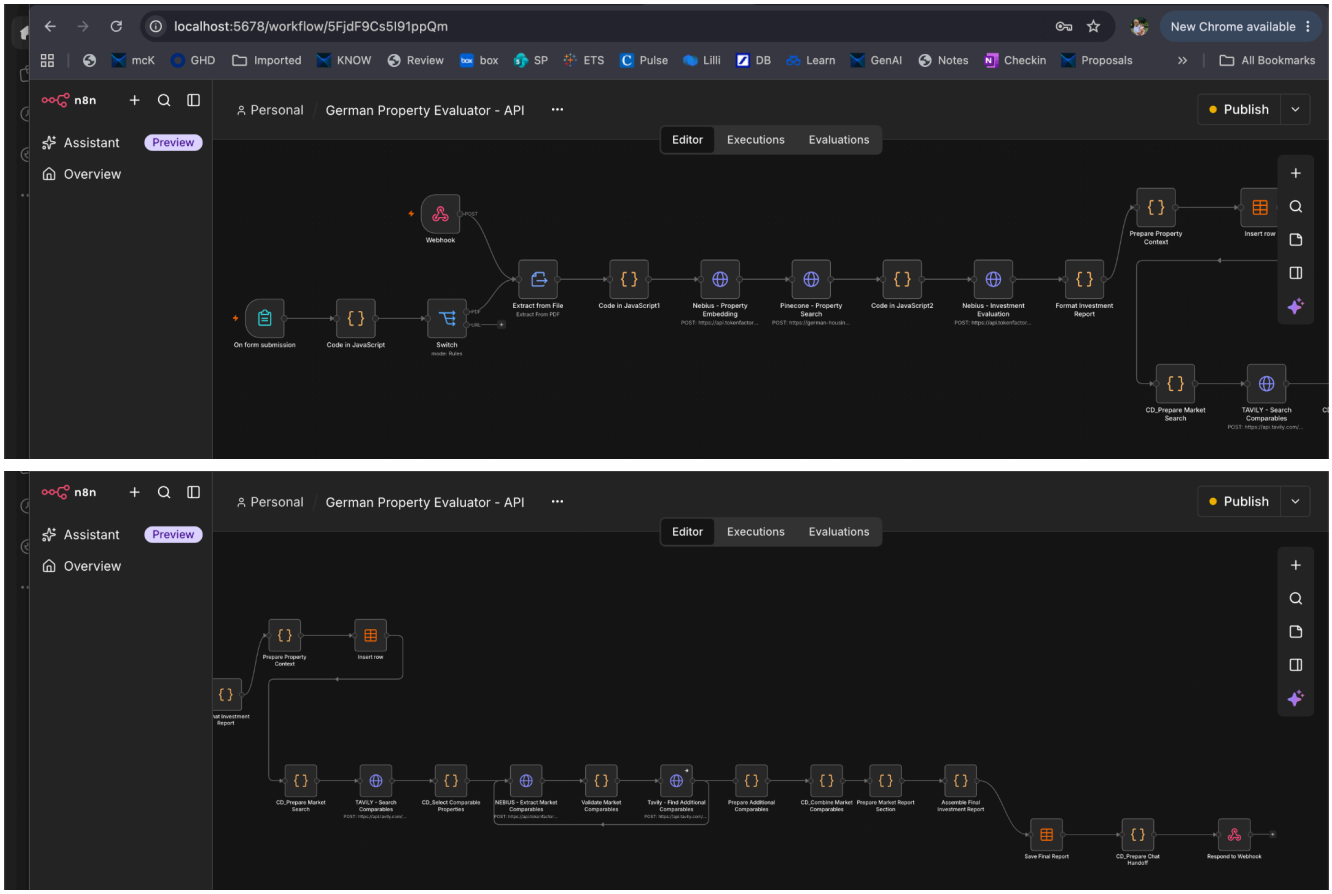


Node in n8n	Description	Key settings
On form submission	Starts the workflow when a user uploads a reference document.	Form: **German Housing Knowledge Base**. Required file field: `document_file`. Accepts `.pdf`, `.xlsx` and `.xls`.
Switch	Routes the uploaded document to the appropriate extraction path.	Checks the file extension: `pdf` → PDF path; `xlsx` or `xls` → Excel path.
Extract from File	Extracts text from a PDF.	Operation: **Extract from PDF**. Input binary field: `document_file`.
Code in JavaScript1	Splits PDF text into smaller, overlapping pieces and attaches source information.	Chunk size: **1,000 characters**; overlap: **200 characters**. Adds a document ID, source filename and chunk index.
Extract from File1	Extracts rows from an Excel spreadsheet.	Operation: **Extract from XLSX**. Input binary field: `document_file`. Header row setting: **off**.
Code in JavaScript	Prepares spreadsheet rows as readable text for embedding.	Searches the first **15 rows** for a likely header; creates a text record and source metadata for each non-empty row.
HTTP Request2	Sends each prepared text piece to Nebius to create an embedding.	Method: **POST**. Model: `Qwen/Qwen3-Embedding-8B`. Input: the prepared text. Uses header-based authentication.
Code in JavaScript2	Matches each embedding with its original text and prepares it for storage.	Checks that the number of responses matches the input and that each vector has **4,096 dimensions**. Retains text, source and document metadata.
HTTP Request1	Stores the embedding and its metadata in Pinecone.	Method: **POST** to the Pinecone upsert endpoint. Namespace: `documents`. Uses header-based authentication.

2 Property evaluation

Input: a property listing PDF. The workflow extracts the listing, retrieves background context, and uses a language model to create a structured investment analysis. It searches for market evidence, assembles an HTML report, saves the property record in an n8n data table, and returns a property ID.

Market evidence is treated carefully. If fewer than three eligible comparable listings are found, the report may show a sourced location benchmark. A broad house or apartment benchmark is labelled as such; unsupported adjustments for property subtype or condition are not assumed.



Major step	Node shown in n8n	Description	Key settings
1. Receive property	**Webhook**	Receives the PDF from the user interface.	**POST** to `/webhook/evaluate-property`; file field: `property_pdf`.
	On form submission	Provides a separate form-based starting point.	Form: **German Property Investment Evaluator**; required PDF field: `property_pdf`.
	Code in JavaScript	Checks the form input and identifies its type.	Requires one input; checks the PDF extension or URL format.
	Switch	Routes form input by type.	Outputs: **PDF** and **URL**. The URL branch is disconnected.
2. Prepare property data	**Extract from File**	Extracts text from the uploaded PDF.	Operation: **Extract from PDF**; binary field: `property_pdf0`.

	Code in JavaScript1	Cleans and prepares the extracted text.	Removes excess whitespace and retains the filename and page count when available.
3. Retrieve knowledge	**Nebius - Property Embedding**	Converts the property text into an embedding for search.	Model: `Qwen/Qwen3-Embedding-8B`.
	Pinecone - Property Search	Finds relevant reference material.	Namespace: `documents`; up to **8 matches** with metadata.
	Code in JavaScript2	Combines the property text with retrieved references.	Prepares reference text, source names and similarity scores.
4. Evaluate property	**Nebius - Investment Evaluation**	Generates the initial investment assessment.	Model: `Qwen/Qwen3-235B-A22B-Instruct-2507`; temperature: **0.2**; maximum **2,500 tokens**.
	Format Investment Report	Validates and formats the assessment.	Parses JSON; checks for an attractiveness score from **1 to 7**; creates report HTML.
	Prepare Property Context	Creates a property record for later use.	Generates a `property_id` and prepares the property text, filename and report JSON.
	Insert row	Saves the initial property record.	Stores the property ID, filename, property text and report JSON in an n8n data table.
5. Research the market	**CD_Prepare Market Search**	Builds a search query from the property details.	Uses available location, property type, area and asking price.
	TAVILY - Search Comparables	Searches for potentially comparable properties.	Search depth: **basic**; maximum **20 results**.
	CD_Select Comparable Properties	Filters and prepares relevant search results.	Uses the subject property's details; does not rely on a fixed city or property type.
	NEBIUS - Extract Market Comparables	Extracts potential comparable properties from search results.	Qwen3 chat model; temperature: **0**; maximum **1,500 tokens**; requests up to **3 candidates**.
	Validate Market Comparables	Checks extracted candidates before they enter the report.	Parses the model response and checks candidates against the subject property and available evidence.

	Tavily - Find Additional Comparables	Performs a further market-price search.	Search depth: **basic**; maximum **8 results**; uses available location, type and condition.
	Prepare Additional Comparables	Processes additional search results into possible market observations.	Looks for location, property type and price information.
	CD_Combine Market Comparables	Combines validated comparables and additional findings.	Handles missing results and removes duplicate or subject-property matches.
6. Complete and return report	**Prepare Market Report Section**	Formats the market comparison.	Uses available comparables, leads and benchmark information; creates HTML.
	Assemble Final Investment Report	Adds the market section to the initial report.	Inserts the comparison into the report HTML and applies table styling.
	Save Final Report	Saves the completed report.	Updates the property's data-table row with `final_report_html`.
	CD_Prepare Chat Handoff	Prepares the completed property record for follow-up chat.	Checks that the property ID and final report are available.
	Respond to Webhook	Returns the result to the interface.	Response format: **JSON**.

3 Property-specific chat

Input: a question, property ID, and session ID. The workflow loads the saved property record and conversation history, then generates a concise answer in English. The IDs keep follow-up questions linked to the selected property and session. The exported chat workflow contains embedding and Pinecone retrieval nodes, but they are not connected to the active chat path. Current chat answers use the saved property context and conversation history; live vector retrieval is a future improvement.

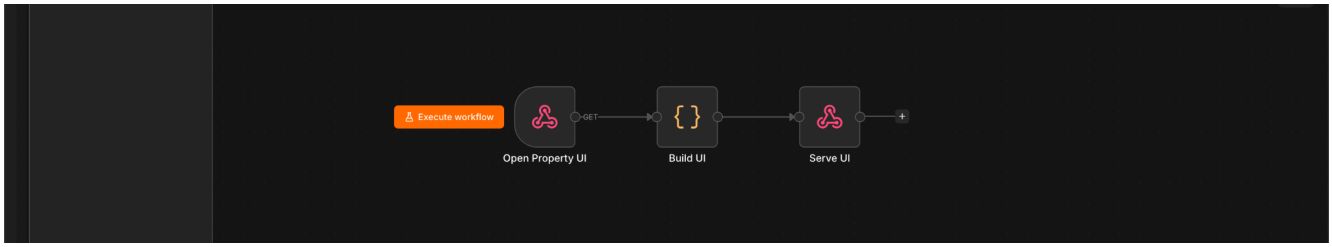


Major step	Node shown in n8n	Description	Key settings
1. Receive question	**Webhook - Property Chat**	Receives a follow-up question from the user interface.	**POST** to `/webhook/property-chat`; response is sent by a later node.
	Prepare Chat Question	Checks and standardizes the incoming request.	Requires `chatInput`, `property_id` and `sessionId`.
2. Load property and history	**Load Property Context**	Retrieves the saved record for the selected property.	Looks up `property_id` in the property evaluation data table; limit: **1 record**.
	Load Chat Memory	Retrieves earlier messages for the same property and conversation.	Filters by both `property_id` and `session_id`.
	Prepare Chat History	Organizes earlier messages for the model.	Sorts messages chronologically and retains the most recent **10 messages**; limits each message to **4,000 characters**.
	Prepare RAG Context	Combines the question, property information, saved evaluation and chat history.	Checks that the requested property matches the loaded record and validates the saved report and score.
3. Generate answer	**Nebius - RAG Chat Answer**	Generates a property-specific answer using the prepared context.	Model: `Qwen/Qwen3-235B-A22B-Instruct-2507`; temperature: **0.2**; maximum **650 tokens**. The prompt instructs the model to answer in **English**.
4. Save conversation	**Prepare Memory Records**	Creates separate records for the user's question and the assistant's answer.	Associates both records with the same `property_id` and `session_id`.
	Save Chat Memory	Saves the question and answer for later follow-up questions.	Writes the session ID, property ID, role and message to the chat history data table.
5. Return answer	**Format Chat Answer**	Checks and prepares the generated answer for the interface.	Returns the answer with the property ID, session ID, question and relevant report details.

	Respond to Webhook	Sends the formatted answer back to the interface.	Response format: **JSON**; response code: **200**.
Disconnected nodes	**When chat message received**	Alternative n8n chat trigger.	Present in the workflow but **not connected** to the active path.
	Nebius - Question Embedding	Would convert a question into an embedding.	Model: `Qwen/Qwen3-Embedding-8B`; the node has **no incoming connection**.
	Pinecone - Chat Retrieval	Would search reference knowledge using the question embedding.	Namespace: `documents`; up to **6 matches**. Its branch is **not connected** to the active chat path.

4 Unified interface

The n8n-hosted interface provides PDF upload, report viewing, and chat on one page. The report is displayed in a restricted frame. Property and session IDs are visible and can be entered again to resume a conversation; the report itself is not restored by that action.



Node shown in n8n	Description	Key settings
Open Property UI	Receives a request to open the property evaluator webpage.	**GET** webhook at `/webhook/property-ui`; uses a response node to return the page.
Build UI	Creates the page, including the PDF upload, report display and chat area. Its JavaScript connects the page to the evaluation and chat workflows.	Uses HTML, CSS and JavaScript. Sends the PDF to `/webhook/evaluate-property` and questions to `/webhook/property-chat`. Keeps the property and session IDs while the page is open.
Serve UI	Returns the completed webpage to the browser.	Responds with HTML text; content type: `text/html; charset=utf-8`; caching: `no-store`.

5. Design Choices

The solution uses a modular n8n architecture to bring document processing, AI analysis, market research and chat into one workflow. Reference documents are indexed for semantic search, while property and session IDs keep reports and follow-up conversations linked to the correct property.

Design choice	Rationale
Four separate n8n workflows	Separates ingestion, evaluation, chat and the UI, making each easier to maintain.
PDF chunking with overlap	Splits reference PDFs into **1,000-character chunks with 200-character overlap** to keep text manageable while preserving context. Spreadsheet rows are prepared as individual text records.
Qwen3-Embedding-8B via Nebius	Converts reference and property text into **4,096-dimensional embeddings** for semantic search.
Pinecone vector index	Stores embeddings with source metadata and retrieves up to **8 relevant reference matches** for property evaluation.
Qwen3-235B-A22B-Instruct-2507 via Nebius	Generates the investment assessment, extracts potential comparables and answers follow-up questions.
Tavily market search	Finds potential comparable properties and broader market-price information to supplement the stored knowledge base.
Property and session IDs	Links saved reports and chat history to the correct property and conversation.
n8n-hosted unified UI	Brings PDF upload, report viewing and chat together on one page, connected to the workflows through webhooks.

6. Lessons and next steps

A. Challenges and lessons learned

- **RAG depends on what was actually indexed.** The evaluator retrieves up to eight Pinecone matches, but having 13 files in the reference folder does not mean all of them were ingested or that every match is relevant.
- **Chunking is a practical design decision.** The ingestion workflow splits PDFs into 1,000-character chunks with 200-character overlap and treats spreadsheet rows as separate records. These choices affect whether useful context is retrieved intact.
- **Different AI models have different roles.** Qwen3-Embedding-8B converts text into vectors for search, while Qwen3-235B generates the investment assessment and chat answers. The embedding model finds related material; it does not assess the property itself.
- **Web search results need validation before the model uses them as evidence.** Tavily can find potential comparable listings, but search snippets may be incomplete or refer to a different property type or location. The workflow therefore filters candidates and distinguishes broader benchmarks from individual comparables.

- **Chat memory is not the same as live retrieval.** The current chat uses the saved property record and recent conversation history. Although embedding and Pinecone nodes exist in the chat workflow, they are disconnected and do not retrieve new reference material for each question.
- **AI output needs checks beyond a successful response.** The evaluator validates the report's JSON and score range, but these checks do not prove that the facts, calculations or market conclusions are correct. Those still need testing against the original documents and sources.

B. Priority improvements

Priority	Next step
High	Connect and test property-filtered vector retrieval in the active chat path.
High	Add webhook authentication and restrict network access; the current webhooks are unauthenticated.
High	Create a repeatable test set for PDF extraction, factual grounding, market evidence, and property isolation.
Medium	Improve progress messages, error handling, report recovery, and measurement of latency and cost.
Medium	Strengthen source verification and clarify how location, property type, condition, and date affect benchmarks.

C. Conclusions

The project demonstrates a working path from property PDF to report and follow-up chat using four connected n8n workflows. Its main value is organizing an initial review in one place. Further testing, stronger security, and more reliable evidence handling are needed before broader use. The report draws on the project's workflow exports and test observations, together with the supplied Customer Support, GraphRAG, Financial Document Analysis, and Enterprise Policy Q&A solution kits.